

Lecture 5: Inference for Subpopulation Means and Differences

September 8, 2023

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- ▶ We also saw that our sample difference in subpopulation proportions was likely to be somewhat close to the population difference in subpopulation proportions.
- ▶ Again, we also demonstrated that we could predict how close we might be by using confidence intervals.
- ▶ We used a bootstrapping approach to confidence intervals and showed how bootstrapping differences allowed us to generate numbers that were close to the numbers in a standard regression table.

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- ▶ Even when the overall sample size is fixed (2246 for our example), different random samples will produce different subgroup sample sizes, and this adds a layer of complexity.

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- ▶ Returning to the CPS example, we will consider 4-year degree holder incomes and income differences between 4-year degree holders and those without a 4-year degree.
- ▶ Even when the overall sample size is fixed (2246 for our example), different random samples will produce different subgroup sample sizes, and this adds a layer of complexity.
- ▶ This complexity will not complicate our analysis much but it does complicate the justification for the analysis.

Subsample and subpopulation means

- A reasonable starting point for describing income in a subsample is its mean income.
- We'll think about the subsample with $X_i = x$ for some value x ,
- e.g. $X_i = 1$ for people with 4-year degrees.
- We can think of our subsample's mean as an estimate of the mean of the corresponding finite subpopulation.

Properties of Population Means: Conditioning

The law of iterated expectations

$$E\{E(Y | X)\} \quad \text{for any random variables } X, Y$$

To calculate the mean of Y , we can average within subpopulations, then across subpopulations.

Irrelevance of independent conditioning variables

$$E(Y | X, X') = E(Y | X) \quad \text{when } X' \text{ is independent of } X, Y$$

If X' is unrelated to X and Y , holding it constant doesn't affect the relationship between them.

Properties of Population Means: Linearity

The linearity of expectations

$$\begin{aligned}E(aY + bZ) &= E(aY) + E(bZ) \\ &= aE(Y) + bE(Z)\end{aligned}$$

for random variables Y, Z and numbers a, b

There are two things going on here.

- To average a sum of two things, we can take two averages and sum.
- To average a constant times a random variable, we can multiply the random variable's average by the constant.
- In other words, we can *distribute* expectations and can pull constants out of them.

The linearity of *conditional* expectations

$$\begin{aligned} E \{ a(X)Y + b(X)Z \mid X \} &= E \{ a(X)Y \mid X \} + E \{ b(X)Z \mid X \} \\ &= a(X)E (Y \mid X) + b(X)E (Z \mid X) \end{aligned}$$

This is like linearity of expectations, but with a twist.

- When we condition on X , we're working with subpopulations in which X is constant.
- This means we can act as if functions of X *are* constants.
- So we can pull them out of expectations that are conditional on X .

It's important to distinguish between two things

1. The *conditional expectation function* $\mu(x) = E[Y | X = x]$.
 - μ is a function; evaluated at x , it's a number. It's not random.
 - It's the mean of the infinite subpopulation of people with $X = x$.
2. The conditional expectation $\mu(X) = E[Y | X]$.
 - $\mu(X)$ is the mean of a *random* subpopulation of people.
 - It's the conditional expectation function *evaluated at* the random variable X .
 - This is the sort of thing that shows up when we use the law of iterated expectations.
 - Let's do an easy calculation to try to make this clear.
 - We'll calculate the mean income in our population using the law of iterated expectations.

Example: Conditioning

- Let's think about a random person drawn from this population
- Y is their income
- X is an indicator for having a four-year degree.
- Suppose the subpopulation means are 100k for people with degrees and 50k for people without.
- And that 7/10 of our population has degrees.

What is $E(Y | X)$?

- $E(Y | X)$ is a random variable that is *either* 100k or 50k
- It's 100k with probability 7/10, when $X = 1$.
- And it's 50k with probability 3/10, when $X = 0$.

What is $E(Y)$?

- It's the mean of this random variable—a number.

$$\begin{aligned} E\{E(Y | X)\} &= E(Y | X = 1)P(X = 1) + E(Y | X = 0)P(X = 0) \\ &= 100k \cdot 7/10 + 50k \cdot 3/10 = 70k + 15k = 85k. \end{aligned}$$

Properties of Population Variances: Conditioning

The law of total variance

$$V[Y] = E\{V(Y | X)\} + V\{E(Y | X)\} \text{ for RVs } X, Y$$

Proof:

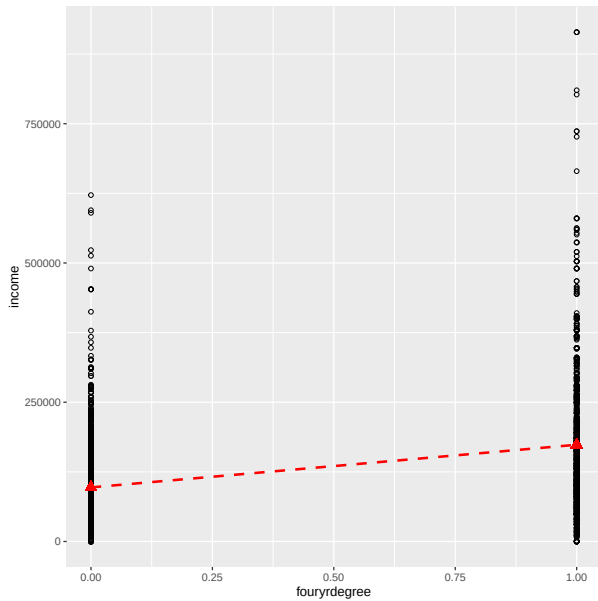
$$\begin{aligned} V[Y] &= E\{Y^2\} - E\{Y\}^2 \\ &= E\{E(Y^2 | X)\} - E\{E(Y | X)\}^2 \\ &= E\{V(Y | X) + E(Y | X)^2\} - E\{E(Y | X)\}^2 \\ &= E\{V(Y | X)\} + [E\{E(Y | X)^2\} - E\{E(Y | X)\}^2] \\ &= E\{V(Y | X)\} + V\{E(Y | X)\} \end{aligned}$$

Income and 4-year degree from the CPS

- ▶ Data on 2022 household income from the CPS for individuals in California aged 25-35

Household	income	4-yr degree
1	250000	1
2	32086	0
⋮	⋮	⋮
11	300640	1
⋮	⋮	⋮
2246	24000	0

CPS Scatterplot



CPS Sampling Distribution

- ▶ Unlike the turnout example, we will never know the true subpopulation income means, which we will denote with μ_1 for those with a 4-yr degree and μ_0 for those without.

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- ▶ We will also never know the subpopulation income standard deviations, which we will denote σ_1 for those with a 4-yr degree and σ_0 for those without.

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- ▶ We will also never know the subpopulation income standard deviations, which we will denote σ_1 for those with a 4-yr degree and σ_0 for those without.
- ▶ We also will not be able to generate a sampling distribution, but we still want to know things about the sampling distribution.

Sampling Distribution for 4-yr degree Mean Income

Let the variable X take the value 1 if Black and 0 if non-Black

i	1	2	...	625		
s	Y_1, X_1	Y_2, X_2	...	Y_{2246}, X_{2246}	$\bar{Y}_{2246}$	$\frac{\sum_{i=1}^{2246} Y_i X_i}{\sum_{i=1}^{2246} X_i}$
1	37,006,0	108,811,1	...	49,001,0	129,514	173,671
2	?,?	?,?	...	?,?	?,?	?,?
3	?,?	?,?	...	?,?	?,?	?,?
1- 4	?,?	?,?	...	?,?	?,?	?,?
5	?,?	?,?	...	?,?	?,?	?,?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	?,?	?,?	...	?,?	?,?	?,?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f(\cdot)$			...		$N(\mu, \frac{\sigma^2}{n})$	?
$E(\cdot)$			...		μ	?
$V(\cdot)$			...		$\frac{\sigma^2}{n}$	?

- ▶ Although we can't observe the population or the sampling distribution, we can mathematically derive certain facts about the sampling distribution.
- ▶ We will focus on the three red question marks.

Property #1: *The sampling distribution of the subsample mean is approximately normal when $\sum_{i=1}^{2246} X_i$ is large.*

- ▶ The approximate normality result comes from a central limit theorem (CLT).
- ▶ We know that $\frac{\sum_{i=1}^{2246} Y_i X_i}{\sum_{i=1}^{2246} X_i} \sim_{approx} N(?, ?)$, but still don't know the mean or the variance of this distribution.

Property #2: *The subsample mean is unbiased for the subpopulation mean.*

Part 1 of Proof (condition on $X_1 = x_1, X_2 = x_2, \dots, X_{2246} = x_{2246}$):

$$\begin{aligned} & E\left[\frac{\sum_{i=1}^{2246} Y_i X_i}{\sum_{i=1}^{2246} X_i} \mid X_1 = x_1, X_2 = x_2, \dots, X_{2246} = x_{2246}\right] \\ &= \frac{1}{\sum_{i=1}^{2246} X_i} E\left[\sum_{i=1}^{2246} Y_i X_i \mid X_1 = x_1, X_2 = x_2, \dots, X_{2246} = x_{2246}\right] \\ &= \frac{1}{\sum_{i=1}^{2246} X_i} \sum_{i=1}^{2246} E[Y_i X_i \mid X_1 = x_1, X_2 = x_2, \dots, X_{2246} = x_{2246}] \\ &= \frac{1}{\sum_{i=1}^{2246} X_i} \sum_{i=1}^{2246} x_i E[Y_i \mid X_1 = x_1, X_2 = x_2, \dots, X_{2246} = x_{2246}] \end{aligned}$$

Property #2: *The subsample mean is unbiased for the subpopulation mean.*

Part 2 of Proof (condition on $X_1 = x_1, X_2 = x_2, \dots, X_{2246} = x_{2246}$):

$$\begin{aligned} & \frac{1}{\sum_{i=1}^{2246} x_i} \sum_{i=1}^{2246} x_i E[Y_i | X_1 = x_1, X_2 = x_2, \dots, X_{2246} = x_{2246}] \\ &= \frac{1}{\sum_{i=1}^{2246} x_i} \sum_{i=1}^{2246} x_i E[Y_i | X_i = x_i] \\ &= \frac{1}{\sum_{i=1}^{2246} x_i} \sum_{i=1}^{2246} x_i [x_i \mu_1 + (1 - x_i) \mu_0] \\ &= \frac{1}{\sum_{i=1}^{2246} x_i} \sum_{i=1}^{2246} x_i \mu_1 \\ &= \frac{1}{\sum_{i=1}^{2246} x_i} \mu_1 \sum_{i=1}^{2246} x_i = \mu_1. \end{aligned}$$

Property #2: *The subsample mean is unbiased for the subpopulation mean.*

Part 3 of Proof:

$$\begin{aligned} & E\left[\frac{\sum_{i=1}^{2246} Y_i X_i}{\sum_{i=1}^{2246} X_i}\right] \\ &= E_{\mathbf{X}}\left[E\left[\frac{\sum_{i=1}^{2246} Y_i X_i}{\sum_{i=1}^{2246} X_i} \mid X_1 = x_1, X_2 = x_2, \dots, X_{2246} = x_{2246}\right]\right] \\ &= E_{\mathbf{X}}[\mu_1] \\ &= \mu_1 \end{aligned}$$

Property #3: The *sampling variance* of the subsample mean,

Part 1 of Proof:

$$\begin{aligned} & V\left[\frac{\sum_{i=1}^{2246} Y_i X_i}{\sum_{i=1}^{2246} X_i}\right] \\ &= E_{\mathbf{X}}\left[V\left[\frac{\sum_{i=1}^{2246} Y_i X_i}{\sum_{i=1}^{2246} X_i} \mid X_1 = x_1, X_2 = x_2, \dots, X_{2246} = x_{2246}\right]\right] \\ &+ V_{\mathbf{X}}\left[E\left[\frac{\sum_{i=1}^{2246} Y_i X_i}{\sum_{i=1}^{2246} X_i} \mid X_1 = x_1, X_2 = x_2, \dots, X_{2246} = x_{2246}\right]\right] \\ &= E_{\mathbf{X}}\left[V\left[\frac{\sum_{i=1}^{2246} Y_i X_i}{\sum_{i=1}^{2246} X_i} \mid X_1 = x_1, X_2 = x_2, \dots, X_{2246} = x_{2246}\right]\right] \\ &+ V_{\mathbf{X}}[\mu_1] \end{aligned}$$

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Property #3: The *sampling variance* of the subsample mean,

Part 3 of Proof:

$$\begin{aligned} & E_{\mathbf{x}}\left[\left(\frac{1}{\sum_{i=1}^{2246} x_i}\right)^2 \sum_{i=1}^{2246} x_i^2 V[Y_i | X_1 = x_1, X_2 = x_2, \dots, X_{2246} = x_{2246}]\right] \\ &= E_{\mathbf{x}}\left[\left(\frac{1}{\sum_{i=1}^{2246} x_i}\right)^2 \sum_{i=1}^{2246} x_i^2 V[Y_i | X_i = x_i]\right] \\ &= E_{\mathbf{x}}\left[\left(\frac{1}{\sum_{i=1}^{2246} x_i}\right)^2 \sum_{i=1}^{2246} x_i^2 (x_i \sigma_1^2 + (1 - x_i) \sigma_0^2)\right] \\ &= E_{\mathbf{x}}\left[\left(\frac{1}{\sum_{i=1}^{2246} x_i}\right)^2 \sum_{i=1}^{2246} x_i^2 \sigma_1^2\right] \\ &= \sigma_1^2 E_{\mathbf{x}}\left[\frac{\sum_{i=1}^{2246} x_i}{(\sum_{i=1}^{2246} x_i)^2}\right] = \sigma_1^2 E_{\mathbf{x}}\left[\frac{1}{\sum_{i=1}^{2246} x_i}\right] \end{aligned}$$

Sampling Distribution for 4-yr degree Mean Income

Let the variable X take the value 1 if 4yr degree and 0 if not.

i	1	2	...	625		
s	Y_1, X_1	...	...	Y_{2246}, X_{2246}	$\bar{Y}_{2246}$	$\frac{\sum_{i=1}^{2246} Y_i X_i}{\sum_{i=1}^{2246} X_i}$
1	37,006,0	...	...	49,001,0	135,406	173,671
2	?,?	...	...	?,?	?,?	?,?
3	?,?	...	...	?,?	?,?	?,?
4	?,?	...	...	?,?	?,?	?,?
5	?,?	...	...	?,?	?,?	?,?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	?,?	...	...	?,?	?,?	?,?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f(\cdot)$		...	...		$N(\mu, \frac{\sigma^2}{n})$	$N(\mu_1, \sigma_1^2 E(\frac{1}{\sum_{i=1}^{2246} X_i}))$
$E(\cdot)$		...	...		μ	μ_1
$V(\cdot)$		...	...		$\frac{\sigma^2}{n}$	$\sigma_1^2 E(\frac{1}{\sum_{i=1}^{2246} X_i}))$

- We still don't know μ_1 , σ_1^2 , or $E(\frac{1}{\sum_{i=1}^{2246} X_i}))$, so we have to bootstrap or estimate them.

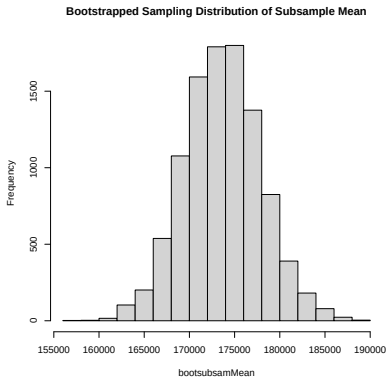
Simulating the Bootstrapped Sampling Distribution

i	1	2	...	2,246	$\bar{y}_{2246}$	$\bar{y}_1 = \frac{\sum_{i=1}^{2246} y_i x_i}{\sum_{i=1}^{2246} x_i}$
y_i	36,006	108,811	...	49,001	129,514	173,671
x_i	0	1	...	0		

Starting with the sample (first row of sampling distribution), each row of the bootstrapped sampling distribution is generated as the following:

```
bootind = sample(1:2246, size=2246, replace=TRUE)
dfboot = dfsam[bootind,]
subsamMean = mean(dfboot$y[dfboot$x == 1])
```

```
B = 10000  
bootsubsamMean = rep(NA,B)  
for( b in 1:B ){  
  bootind = sample(1:2246,size=2246,replace=T)  
  dfboot = dfsam[bootind,]  
  bootsubsamMean[b] = mean(dfboot$y[dfboot$x==1])  
}
```



Normal Approximation CI

Estimates

- ▶ $\mu_1: \bar{y}_1 = \frac{\sum_{i=1}^{2246} y_i x_i}{\sum_{i=1}^{2246} x_i}$
- ▶ $\sigma_1^2: \frac{\sum_{i=1}^{2246} (y_i - \bar{y}_1)^2 x_i}{\sum_{i=1}^{2246} x_i - 1}$
- ▶ $E\left(\frac{1}{\sum_{i=1}^{2246} x_i}\right): \frac{1}{\sum_{i=1}^{2246} x_i}$

95% CI:

$$\bar{y}_1 \pm 1.96 \cdot \sqrt{\frac{1}{\sum_{i=1}^{2246} x_i} \cdot \frac{\sum_{i=1}^{2246} (y_i - \bar{y}_1)^2 x_i}{\sum_{i=1}^{2246} x_i - 1}}$$

95% CIs

- ▶ bootstrapping: [\$165,520, \$182,219]
- ▶ approximation: [\$165411, \$181930]

Sampling Distribution for Difference in Mean Income

Let the variable X take the value 1 if 4yr degree and 0 if not

i	1	2	...	625		
s	Y_1, X_1	...	...	Y_{2246}, X_{2246}	$\frac{\sum_{i=1}^{2246} Y_i X_i}{\sum_{i=1}^{2246} X_i}$	$\frac{\sum_{i=1}^{2246} Y_i X_i}{\sum_{i=1}^{2246} X_i} - \frac{\sum_{i=1}^{2246} Y_i (1-X_i)}{\sum_{i=1}^{2246} (1-X_i)}$
1	37,006,0	...	...	49,001,0	173,671	76,583
2	?,?	...	...	?,?	?,?	?,?
3	?,?	...	...	?,?	?,?	?,?
4	?,?	...	...	?,?	?,?	?,?
5	?,?	...	...	?,?	?,?	?,?
...	...	...	...	...	...	...
1 mil	?,?	...	...	?,?	?,?	?,?
...	...	...	...	...	...	...
$f(\cdot)$		...	...		$N(\mu, \frac{\sigma^2}{n})$	$N(\cdot, \cdot)$
$E(\cdot)$		...	...		μ	$\mu_1 - \mu_0$
$V(\cdot)$		...	...		$\frac{\sigma^2}{n}$	$\sigma_1^2 E(\frac{1}{\sum_{i=1}^{2246} X_i}) + \sigma_0^2 E(\frac{1}{\sum_{i=1}^{2246} (1-X_i)})$

- We still don't know $\mu_1, \mu_0, \sigma_1^2, \sigma_0^2, E(\frac{1}{\sum_{i=1}^{2246} X_i})$, or $E(\frac{1}{\sum_{i=1}^{2246} (1-X_i)})$, so we have to bootstrap or estimate them.

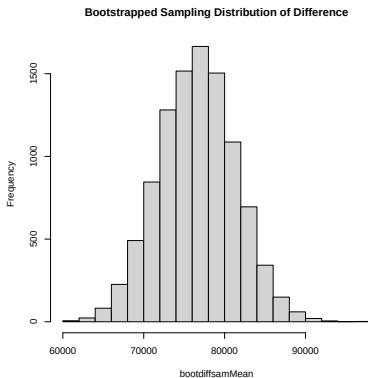
Simulating the Bootstrapped Sampling Distribution

i	1	2	...	2,246	$\frac{\sum_{i=1}^{2246} y_i x_i}{\sum_{i=1}^{2246} x_i} - \frac{\sum_{i=1}^{2246} y_i (1-x_i)}{\sum_{i=1}^{2246} (1-x_i)}$
y_i	36,006	108,811	...	49,001	76,583
x_i	0	1	...	0	

Starting with the sample (first row of sampling distribution), each row of the bootstrapped sampling distribution is generated as the following:

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bootind = sample(1:2246, size=2246, replace=TRUE)
dfboot = dfsam[bootind,]
bootdiffsamMean = mean(dfboot$y[dfboot$x == 1]) -
mean(dfboot$y[dfboot$x == 0])
```

```
B = 10000
bootdiffsamMean = rep(NA,B)
for( b in 1:B ){
bootind = sample(1:2246,size=2246,replace=T)
dfboot = dfsam[bootind,]
bootdiffsamMean[b] = mean(dfboot$y[dfboot$x==1]) - mean(
}
```



Normal Approximation CI

Estimates

- ▶ $\mu_1 - \mu_0: \bar{y}_1 - \bar{y}_0$
- ▶ $\sigma_1^2: \frac{\sum_{i=1}^{2246} (y_i - \bar{y}_1)^2 x_i}{\sum_{i=1}^{2246} x_i - 1}$
- ▶ $E\left(\frac{1}{\sum_{i=1}^{2246} x_i}\right): \frac{1}{\sum_{i=1}^{2246} x_i}$
- ▶ $\sigma_0^2: \frac{\sum_{i=1}^{2246} (y_i - \bar{y}_1)^2 (1 - x_i)}{\sum_{i=1}^{2246} (1 - x_i) - 1}$
- ▶ $E\left(\frac{1}{\sum_{i=1}^{2246} (1 - x_i)}\right): \frac{1}{\sum_{i=1}^{2246} (1 - x_i)}$

95% CI:

$$\bar{y}_1 - \bar{y}_0 \pm 1.96 \cdot \sqrt{\frac{1}{\sum_{i=1}^{2246} x_i} \cdot \frac{\sum_{i=1}^{2246} (y_i - \bar{y}_1)^2 x_i}{\sum_{i=1}^{2246} x_i - 1} + \frac{1}{\sum_{i=1}^{2246} x_i} \cdot \frac{\sum_{i=1}^{2246} (y_i - \bar{y}_1)^2 (1 - x_i)}{\sum_{i=1}^{2246} (1 - x_i) - 1}}$$

95% CIs for Difference

- ▶ bootstrapping: [\$67,438, \$85,865]
- ▶ approximation: [\$65,705, \$87,459]